Shayan Jalali

+39 351 339 2593 · shayanjalali.bioinformatics@gmail.com · Novara, Italy

LinkedIn: linkedin.com/in/shayanjl · GitHub: github.com/Shayanjl · Website: shayanjl.github.io · ORCID: 0009-0005-3909-2920

Research Profile

MSc graduate in Medical Biotechnologies from Università del Piemonte Orientale, Italy, with thesis research completed through an Erasmus+ placement at the Royal College of Surgeons in Ireland (RCSI), Dublin. My research interests include computational systems biology, particularly modelling AML-driven disruption of erythropoiesis using Julia, with a focus on ODE-based biological modelling and global sensitivity analysis. I have experience in bioinformatics, systems biology, transcriptomics, genomics-driven biomarker discovery, machine learning, and laboratory work across Ireland, Italy, and Iran. My background includes international research and laboratory experience in computational biology at RCSI, systems biology and cancer data analysis at UPO-CAAD, COVID-19 PCR diagnostics, and plant biotechnology research in Iran. I am particularly interested in quantitative systems biology, computational biology, single-cell and omics-informed modelling of biological systems, translational bioinformatics, and emerging applications of radiomics in translational imaging. I have co-authored several scholarly outputs, including first-author work, in areas such as transcriptomics, biomarker discovery, gastric cancer prediction, AML niche modelling, biotechnology, artificial intelligence applications, genomics, and computational biology. I am proficient in Python, R, Julia, Bash, Git, MATLAB, LaTeX, BibTeX, Overleaf, EndNote, Zotero, Cytoscape, Galaxy, GEO2R, RapidMiner, XMind, and a range of omics and systems biology tools.

Education

MSc in Medical Biotechnologies (Oct 2023 – Mar 2026), Università del Piemonte Orientale (UPO), Novara, Italy

- Thesis: Mathematical Modelling and Sensitivity Analysis of Nurse Macrophage-Driven Erythropoiesis Disruption in Acute Myeloid Leukaemia (Abstract | Full text)
- Thesis supervisor: Dr. Marc Sturrock, Royal College of Surgeons in Ireland (RCSI), Department of Biophysics & Systems Biology, Dublin, Ireland
- Final grade: 105/110

BSc in Cellular and Molecular Biology – Biotechnology (2019 – 2023), IAU University, Ahvaz, Iran

- Final grade: Very Good

Research Experience

Visiting Scientist, Department of Biophysics & Systems Biology — Royal College of Surgeons in Ireland (RCSI), Dublin, Ireland (Feb 2025 – Sep 2025)

- Conducted Master's thesis research at RCSI under the supervision of Dr. Marc Sturrock
- Investigated how acute myeloid leukaemia disrupts erythroblastic islands and nurse macrophages during haematopoiesis
- Modelled AML-induced disruption of erythroblastic islands and nurse macrophages using ordinary differential equations in Julia
- Developed ODE-based models in Julia using DifferentialEquations.jl
- Developed toy models to support global sensitivity analysis and method evaluation
- Implemented and compared global sensitivity analysis methods, including Sobol and Morris, to identify five key regulators of impaired erythropoiesis
- Ran simulation-based analyses of intervention strategies affecting macrophage and erythroid dynamics
- Conducted more than 50 simulations to evaluate M-CSF treatment strategies
- Explored regulatory thresholds influencing macrophage dynamics
- Contributed to manuscript preparation, data interpretation, and scientific discussion
- Synthesised literature from more than 100 references and integrated biological evidence to support model development
- Analysed multi-source biological datasets to support model development and interpretation
- Collaborated within an interdisciplinary research team

- Presented research findings at departmental seminars and weekly meetings, and prepared scientific posters and professional slide presentations
- Co-authored a manuscript on AML-associated disruption of bone marrow erythropoiesis, currently under review

Graduate Teaching Assistant & Research Trainee in Systems Biology — Università del Piemonte Orientale (UPO) — CAAD Center, Novara, Italy (Oct 2023 — Feb 2025)

- Performed transcriptomic and biomarker analyses in R and Python across public cancer datasets, including GEO and TCGA
- Contributed to computational workflows for biomarker discovery, pathway analysis, and biological interpretation
- Analysed 10+ transcriptomic datasets and validated biomarkers across GEO and TCGA
- Used systems biology and machine learning approaches to identify potential biomarkers and study molecular mechanisms relevant to treatment strategies
- Supported teaching activities in computational modelling and data analysis for undergraduate and postgraduate students
- Developed and delivered tutorials for undergraduate and postgraduate students on computational modelling and data analysis
- Contributed to the design of the *Real-Time Radiomics for Paediatric Brain Tumour Triage* proposal, focusing on scanner harmonisation, clinical workflow integration, and AI-based radiomics for diagnostic triage
- Contributed to a machine learning study on peritoneal metastasis prediction in gastric cancer, using R for statistical analysis, feature evaluation, and comparison of six machine learning models across clinical variables including lymph node status, invasion depth, and inflammatory markers

Laboratory Assistant — PCR Diagnostics (COVID-19 Response) — Golestan Hospital, Tehran, Iran (2022 — 2023)

- Volunteered at the PCR Laboratory during the COVID-19 pandemic
- Assisted with PCR-based diagnostics
- Participated in sample preparation
- Supported workflow organisation
- Worked under biosafety compliance procedures
- Gained experience in molecular diagnostics, teamwork, and crisis response

Intern — Plant Biotechnology Laboratory (Sugarcane Biotechnology Research) — Iranian Sugarcane and By-Products Research and Training Institute (ISCRTI), Ahvaz, Iran (Sep 2020 — Mar 2021)

- Analysed molecular markers and genetic variation in sugarcane using gamma mutagenesis to support development of higher-yield varieties
- Maintained sterile culture environments using agar-based jars
- Operated laboratory instruments including centrifuges, microscopes, and autoclaves
- Developed skills in experimental design, data interpretation, and scientific reporting within an interdisciplinary research setting

Technical Skills

Programming and Software

- Python | R | Julia | MATLAB | Bash | Git | GitBash | GitHub | Overleaf | LaTeX | BibTeX | EndNote | Zotero | Cytoscape | Galaxy | GEO2R | RapidMiner | Microsoft Excel | XMind

R Packages, Libraries, and Analytical Tools

- tidyverse | caret | ggplot2 | plotly | dplyr | limma | DESeq2 | WGCNA | clusterProfiler

Data Resources and Platforms

- GEO | TCGA | ArrayExpress | UniProt

Bioinformatics and Data Analysis

- Bulk RNA-seq analysis | RNA-seq | basic single-cell RNA-seq analysis | Transcriptomic analysis | Transcriptomic and genomic data integration | Multi-omics integration | Microarray analysis | Differential expression analysis | Pathway enrichment analysis | Biomarker discovery | Disease modelling | External dataset validation using GEO | Data integration | Data cleaning | Data normalisation | Feature selection | Statistical testing | Clinical data interpretation

Feature Selection and Statistics

- Gini Index | Relief | Information Gain | t-tests | Chi-squared tests | Correlation analysis

Systems Biology and Modelling

- Biostatistics | ODE modelling in Julia | Ordinary differential equation modelling | Data-driven biological modelling | Predictive modelling | Model development and validation | Model tuning
- Cross-validation | Performance evaluation | Parameter optimisation | Parameter inference for biological models | Simulation | Simulation-based analysis | Global sensitivity analysis | Sobol | Morris

Machine Learning

- Logistic Regression | Random Forest | Support Vector Machines | K-Nearest Neighbors | Naïve Bayes | Decision Trees | Supervised learning | Machine learning for biomarker discovery and prediction modelling

Omics and Pathway Tools

- Enrichr | Reactome | GSEA | miRTarBase

Radiomics-Related Familiarity

- Core radiomics concepts | Image-based feature extraction | Harmonisation methods including ComBat and CovBat | Integration into AI-driven diagnostic frameworks

Scientific Writing and Research Communication

- Academic writing | Literature review | Manuscript preparation | Research proposal writing | Scientific publication writing | Reference management | Public speaking | Teaching

Lab and Clinical Skills

- DNA extraction | PCR | Flow cytometry | Immunohistochemistry | Tissue culture, including plant and human cells | Clinical data interpretation | Molecular diagnostics | Sample preparation | Biosafety compliance

Publications & Scholarly Outputs

- Mosteo, L., Rocha, L., Mai, C., Maceiras, A. R., Cavadas, B., Pereira, C., Lopes, M., Chacim, S., Ramos, I., Hawkins, E., Mariz, J. M., Porto, G., **Jalali, S.**, Sturrock, M., Lo Celso, C., & Duarte, D. (2025). *Remodeling of nurse macrophages and erythroblastic islands contributes to the loss of bone marrow erythropoiesis in acute myeloid leukemia*. Manuscript under review
- Hassani, S., Momeni, M., Siahtiri, M., Massahi, S., **Jalali, S.**, Salehi, S., & Gholaminejad, A. (2025). *From tumor pathways to blood signature: A machine-learning-validated 5-miRNA signature for the early detection of gastric cancer*. Preprints.org. https://www.preprints.org/manuscript/202510.1162. Preprint under review
- **Jalali, S.**, Dadkhah, K., & Mirsaeid Ghazi, M. (2025). *Peritoneal metastasis prediction in gastric cancer: A machine learning approach*. medRxiv. https://doi.org/10.1101/2025.04.11.25325702. Preprint under review
- **Jalali, S.** (2023). *Investigating the use of nanobodies and the use of biotechnology methods in cancer treatment.* Presented and published in the 1st International Congress of Cancer Genomics, Tehran, Iran, May 3-5, 2023. Available at: https://civilica.com/doc/1823184/
- Jabbarzadeh Kaboli, P., **Jalali, S.**, Mohammadalizadeh Rami, S., & Salahi, S. (2026). *BioLionaire Scientist: Volume 6 – Code Empire*. Co-authored volume on computational biology, bioinformatics, genomics, artificial intelligence applications, and biotechnology innovation and business strategy. Published by OncoVanguard Media, LLC

Selected Courses & Certifications

- Mastering Git, GitHub, and GitLab: Your Ultimate Guide to Version Control | Artificial Intelligence Foundations | Foundations of Working with Database in Python | Julia Scientific Programming | Fundamentals of R | Machine Learning | Supervised Machine Learning: Regression and Classification | Regenerative Medicine | Data Mining | Regression | Anatomy of the Chest, Neck, Abdomen, and Pelvis | The Science of Stem Cells | Drug Commercialization | Drug Development | Drug Discovery | Genes and the Human Condition (From Behavior to Biotechnology) | Industrial Biotechnology

Scholarships & Awards

- Erasmus+ Research Mobility Award, Royal College of Surgeons in Ireland (RCSI), 2025
- Italian Government Master's Scholarship, Università del Piemonte Orientale, 2023–2025

Languages

- English: C1 (reading, writing, listening, and speaking); MSc studies in Italy and Erasmus+ research at RCSI conducted in English

Academic Reference

- Available upon request